

Immanuel S S

[Github](#) | [Portfolio Website](#)

Location: Dubai,UAE

Email: s.s.immanuel149@gmail.com

Backend and systems-focused software engineer with hands-on experience across robotics, computer vision, and cross-platform application development. Comfortable working in C++, Python, and Rust, with a strong interest in low-latency systems, distributed architectures, and networked applications. Seeking a software engineering role where I can apply this background to build reliable, performant backend systems. Also maintains a side interest in NLP, LLMs, and AR/VR.

TECHNICAL SKILLS

Languages : C, C++, Rust, Python, SQL, Java, assembly, Kotlin, JavaScript
Frameworks : ROS2, Tauri, LangGraph, PyTorch, React
Systems : Linux, Docker, Git, WebSockets/TLS, SQLite
Dev Tools : Visual Studio, Rider, Visual Studio Code, Android Studio

PROJECTS

ROS2 Agentic Fetch Robot

[Source Code](#)

- Developed an agentic AI system that interprets natural-language voice commands and autonomously orchestrates a ROS2 mobile manipulator to locate and retrieve objects in a Gazebo simulation.
- Integrated a YOLO11 object detection pipeline with monocular camera-based 3D localization (no depth camera), feeding detections into a Groq Llama-3.3-70B LLM agent (LangGraph) that dynamically plans and executes multi-step pick-and-place tasks under an event-driven control loop.
- Implemented SLAM-based autonomous mapping and fault-tolerant navigation using a bisect-retry strategy — recursively halving the path on failure to preserve partial progress — built on a modular ROS2 architecture with Nav2 and MoveIt2 designed for portability across simulation environments.

On Screen Translator

[Source Code](#)

- Built a real-time on-screen Korean-to-English translator that captures the display via DXcam (60 FPS), runs EasyOCR text detection/recognition on a CUDA GPU, and renders translated text as a transparent overlay aligned directly on top of the source window — entirely offline, no cloud APIs.
- Designed a producer-consumer pipeline with a detector thread, IoU-based frame-to-frame association, and a bounded OCR/translation queue; introduced a content-keyed translation cache that eliminates redundant model calls and keeps the hot path at detection speed.
- Integrated a locally hosted, quantized instruction-tuned LLM for context-aware translation of domain-specific terminology (wuxia/webnovel vocabulary), co-locating OCR and LLM inference on a single consumer GPU and resolving ambiguities that off-the-shelf neural MT systems mistranslate.

Gallery – Cross-Platform LAN Media Backup

Source pending public release

- Built a cross-platform photo/video gallery that syncs an Android phone and Windows desktop entirely over the local network, with zero cloud relay or third-party servers, pairing the two devices via a QR-coded, TLS-pinned wss:// handshake.
- Designed a SHA-256 content-hash catalog protocol that reconciles renamed and duplicate files by content rather than filename, syncing a paginated, delta-aware media index and thumbnails between two independent native clients (Kotlin/Jetpack Compose on Android; Rust/Tauri with axum, rustls, and tokio on desktop).
- Implemented an additive-only backup model where full-resolution originals transfer only on explicit request (single push or PC-side multi-select batch over a live-connection registry), guaranteeing phone-side deletions never remove already-backed-up files on the PC.

EDUCATION

Birla Institute of Technology and Science Pilani
Bachelor of Engineering Computer Science

Dubai, UAE
Sep 2022 – Aug 2026

ACHIEVEMENTS

- B1 Delf

EXPERIENCE

White Paper Alliance IT Solutions

AI/ML Intern

Dubai, UAE

August 2025 - January 2026

- Reduced WhatsApp Chatbot agent latency by 50% by optimizing backend workflows.
- Built and debugged 5-language voice-to-text support, and handled testing + production fixes for chatbot-agent integration.
- Built a sports RAG chatbot using SQL and LangGraph, handling data collection, preprocessing, and schema design.
- Developed and integrated SQL-based retrieval workflows; later migrated to MongoDB and implemented a new RAG pipeline for NoSQL.
- Created a Streamlit demo UI and contributed to score projection logic for analytics features.
- Developed an ML-based video processing pipeline to perform seamless image overlay using object detection models.
- Collected and curated training data, and trained/tuned models to improve detection accuracy and overlay quality.
- Optimized model performance for more stable and realistic video outputs.
- Built fake news detection models using DistilBERT and regression techniques.
- Performed data preprocessing, training, and evaluation to enhance prediction accuracy.
- Implemented real-time face tracking using YOLO-based object detection.
- Developed hand tracking system using YOLO and MediaPipe for gesture/position detection.

Unique World Robotics

Intern

Dubai, UAE

June 2024 - August 2024

- Led and actively participated in hands-on sessions using Arduino, Cretile, ESP32, and LEGO EV3, guiding 20 participants per session through automation and microcontroller-based projects.
- Designed and demonstrated 6 projects per day, incorporating sensor integration, basic automation, and microcontroller programming, simplifying complex STEM concepts into engaging, beginner-friendly formats.
- Presented smart home circuit ideas using Cretile kits (e.g., motion-sensor lights, temperature triggers), helping participants understand how fundamental circuitry maps to real-world IoT applications.
- Oversaw planning and execution of workshops, managing logistics, resource allocation, and session flow, while ensuring smooth hands-on experiences.
- Provided individual technical mentoring and troubleshooting support, balancing leadership with collaborative learning across a 6-member team.
- Gained practical familiarity with hardware setup, pin configurations, and real-world constraints of embedded systems.
- Received highly positive feedback from over 95% of participants, with many expressing satisfaction and enthusiasm for the sessions and projects.